SIMPLE MAD MASTER-PATTERN EXAMPLE

Student Registration → Home

This is a deliberately small example for understanding the pattern. It is not meant to demonstrate every feature from the big master program.

Pattern used: Activity 1 → form → validation → RadioButton + CheckBox → SharedPreferences → AlertDialog → Bundle → Explicit Intent → Activity 2 → ListView + ArrayAdapter → click → AlertDialog/Toast → long press → Context Menu → Implicit Intent.

Exam idea: Once you understand this program, replace Student Name / Course / Hobbies / list items / messages / Intent destination with whatever the question asks.

1. QUESTION FOR THIS EXAMPLE

Write an Android application to register a student. Take student name and course, select gender using RadioButton, select hobbies using CheckBoxes, validate the name, save the name using SharedPreferences, and move to a second Activity using an Explicit Intent. In the second Activity, display a ListView of subjects. On normal click show an AlertDialog and Toast. On long press show a Context Menu with Delete and Open Website.

This single small question lets you practice the reusable structure without making the program unnecessarily large.

2. PROJECT FILES

```
SimpleMAD/
└── app/src/main/
    ├── java/com/example/simplemad/
    │   ├── MainActivity.java
    │   └── HomeActivity.java
    ├── res/layout/
    │   ├── activity_main.xml
    │   └── activity_home.xml
    ├── res/menu/
    │   └── context_menu.xml
    └── AndroidManifest.xml
```

3. ACTIVITY 1 XML — activity_main.xml

Keep the XML simple. The important thing is to understand the IDs because Java uses these IDs with findViewById().

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="20dp">

        <!-- Student name -->
        <EditText
            android:id="@+id/nameEditText"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Student Name"
            android:inputType="textPersonName" />

        <!-- Course -->
        <EditText
            android:id="@+id/courseEditText"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Course"
            android:inputType="text" />

        <!-- Only one gender can be selected -->
        <RadioGroup
            android:id="@+id/genderRadioGroup"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="horizontal">

            <RadioButton
                android:id="@+id/maleRadioButton"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Male" />

            <RadioButton
                android:id="@+id/femaleRadioButton"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Female" />
        </RadioGroup>

        <!-- More than one hobby can be selected -->
        <CheckBox
            android:id="@+id/readingCheckBox"
            android:layout_width="wrap_content"
```

```
            android:layout_height="wrap_content"
            android:text="Reading" />

        <CheckBox
            android:id="@+id/sportsCheckBox"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Sports" />

        <Button
            android:id="@+id/registerButton"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Register" />

        <Button
            android:id="@+id/showSavedButton"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Show Saved Name" />

    </LinearLayout>
</ScrollView>
```

4. ACTIVITY 1 JAVA — MainActivity.java

Read this slowly. The comments show what each section is doing and why it exists.

```
package com.example.simplemad;

import android.app.AlertDialog;
import android.content.Intent;
import android.content.SharedPreferences;
import android.os.Bundle;
import android.view.View;
import android.widget.*;

public class MainActivity extends Activity {

    // Variables for the XML widgets
    EditText nameEditText;
    EditText courseEditText;

    RadioGroup genderRadioGroup;
    CheckBox readingCheckBox;
    CheckBox sportsCheckBox;

    Button registerButton;
    Button showSavedButton;

    // SharedPreferences stores small data permanently
    SharedPreferences preferences;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        // Connect Java with activity_main.xml
        setContentView(R.layout.activity_main);

        // Find each XML view using its ID
        nameEditText = findViewById(R.id.nameEditText);
        courseEditText = findViewById(R.id.courseEditText);

        genderRadioGroup = findViewById(R.id.genderRadioGroup);
        readingCheckBox = findViewById(R.id.readingCheckBox);
        sportsCheckBox = findViewById(R.id.sportsCheckBox);

        registerButton = findViewById(R.id.registerButton);
        showSavedButton = findViewById(R.id.showSavedButton);

        // Create/open a SharedPreferences file
        preferences = getSharedPreferences(
                "StudentPrefs",
                MODE_PRIVATE
        );

        // Register button click
        registerButton.setOnClickListener(v -> registerStudent());

        // Show saved name button click
        showSavedButton.setOnClickListener(
                v -> showSavedName()
        );
    }

    // ------------------------------------------------
    // VALIDATION + FORM READING + SHARED PREFERENCES
    // ------------------------------------------------

    private void registerStudent() {

        // Read text from EditTexts
        String name = nameEditText
                .getText()
                .toString()
                .trim();

        String course = courseEditText
                .getText()
                .toString()
                .trim();

        // Simple validation
        if (name.isEmpty()) {
            nameEditText.setError("Enter student name");
            nameEditText.requestFocus();
            return;
        }

        if (course.isEmpty()) {
```

```
            courseEditText.setError("Enter course");
            courseEditText.requestFocus();
            return;
        }

        // Read selected RadioButton
        int selectedId =
                genderRadioGroup.getCheckedRadioButtonId();

        String gender = "Not selected";

        if (selectedId != -1) {
            RadioButton selectedRadioButton =
                    findViewById(selectedId);

            gender = selectedRadioButton
                    .getText()
                    .toString();
        }

        // Read CheckBox values
        String hobbies = "";

        if (readingCheckBox.isChecked()) {
            hobbies += "Reading ";
        }

        if (sportsCheckBox.isChecked()) {
            hobbies += "Sports";
        }

        if (hobbies.isEmpty()) {
            hobbies = "None";
        }

        // Save the name in SharedPreferences
        preferences.edit()
                .putString("studentName", name)
                .apply();

        // AlertDialog gives a confirmation
        new AlertDialog.Builder(this)
                .setTitle("Registration")
                .setMessage(
                        "Name: " + name +
                        "\nCourse: " + course +
                        "\nGender: " + gender +
                        "\nHobbies: " + hobbies
                )
                .setPositiveButton("Continue",
                        (dialog, which) -> openHome(name))
                .setNegativeButton("Cancel", null)
                .show();
    }

    // ------------------------------------------------
    // EXPLICIT INTENT + BUNDLE
    // ------------------------------------------------

    private void openHome(String name) {

        // Explicit Intent means we directly specify
        // which Activity should open.
        Intent intent = new Intent(
                MainActivity.this,
                HomeActivity.class
        );

        // Bundle stores data to send to the next Activity
        Bundle bundle = new Bundle();
        bundle.putString("studentName", name);

        // Put Bundle inside Intent
        intent.putExtras(bundle);

        // Start the second Activity
        startActivity(intent);
    }

    // ------------------------------------------------
    // READ SHARED PREFERENCES
    // ------------------------------------------------

    private void showSavedName() {

        // Read the saved value.
        // "No name saved" is the default value.
        String savedName = preferences.getString(
```

```
                "studentName",
                "No name saved"
        );

        Toast.makeText(
                this,
                "Saved: " + savedName,
                Toast.LENGTH_SHORT
        ).show();
    }
}
```

5. WHAT TO UNDERSTAND FROM MainActivity

Code	Concept
findViewById()	Connects Java with XML views.
getText().toString().trim()	Reads text from EditText.
setError()	Shows validation error.
RadioGroup	Allows one RadioButton choice.
isChecked()	Checks a CheckBox.
SharedPreferences	Stores small persistent data.
AlertDialog	Shows confirmation/details.
Bundle	Carries data between components.
Explicit Intent	Opens a specific Activity.

6. ACTIVITY 2 XML — activity_home.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:id="@+id/welcomeTextView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="22sp"
        android:textStyle="bold" />

    <!-- ListView displays an array/list of subjects -->
    <ListView
        android:id="@+id/subjectListView"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1" />

</LinearLayout>
```

7. ACTIVITY 2 JAVA — HomeActivity.java

```
package com.example.simplemad;

import android.app.*;
import android.content.*;
import android.net.Uri;
import android.os.Bundle;
import android.view.*;
import android.widget.*;
import java.util.ArrayList;

public class HomeActivity extends Activity {

    TextView welcomeTextView;
    ListView subjectListView;

    ArrayList<String> subjects;
    ArrayAdapter<String> adapter;

    int selectedPosition = -1;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_home);

        welcomeTextView =
                findViewById(R.id.welcomeTextView);

        subjectListView =
                findViewById(R.id.subjectListView);

        // -----------------------------------------
        // READ DATA FROM EXPLICIT INTENT / BUNDLE
        // -----------------------------------------

        Bundle bundle = getIntent().getExtras();

        if (bundle != null) {

            String name = bundle.getString(
                    "studentName",
                    "Student"
            );

            welcomeTextView.setText(
                    "Welcome, " + name
            );
        }

        // -----------------------------------------
        // LISTVIEW + ARRAYADAPTER
        // -----------------------------------------

        subjects = new ArrayList<>();

        subjects.add("Java");
        subjects.add("Android");
        subjects.add("XML");
```

```
        subjects.add("Mobile Computing");

        // Adapter connects the ArrayList to ListView
        adapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_list_item_1,
                subjects
        );

        subjectListView.setAdapter(adapter);

        // -----------------------------------------
        // NORMAL CLICK
        // -----------------------------------------

        subjectListView.setOnItemClickListener(
                (parent, view, position, id) -> {

                    String subject =
                            subjects.get(position);

                    // Show selected subject
                    new AlertDialog.Builder(this)
                            .setTitle("Subject Selected")
                            .setMessage(subject)
                            .setPositiveButton("OK", null)
                            .show();

                    // Also show Toast
                    Toast.makeText(
                            this,
                            "You selected " + subject,
                            Toast.LENGTH_SHORT
                    ).show();
                }
        );

        // -----------------------------------------
        // LONG CLICK
        // -----------------------------------------

        subjectListView.setOnItemLongClickListener(
                (parent, view, position, id) -> {

                    // Remember which item was long pressed
                    selectedPosition = position;

                    // Show Context Menu
                    subjectListView.showContextMenu();

                    return true;
                }
        );

        // Context menu is attached to the ListView
        registerForContextMenu(subjectListView);
    }

    // -----------------------------------------
    // CREATE CONTEXT MENU
    // -----------------------------------------

    @Override
    public void onCreateContextMenu(
            ContextMenu menu,
            View v,
            ContextMenu.ContextMenuInfo menuInfo) {

        super.onCreateContextMenu(
                menu,
                v,
                menuInfo
        );

        menu.setHeaderTitle("Choose Action");

        menu.add("Delete");
        menu.add("Open Website");
    }

    // -----------------------------------------
    // CONTEXT MENU CLICK
    // -----------------------------------------

    @Override
    public boolean onContextItemSelected(
            MenuItem item) {

        if (selectedPosition == -1) {
```

```
            return super.onContextItemSelected(item);
        }

        String selectedSubject =
                subjects.get(selectedPosition);

        // If Delete was selected
        if (item.getTitle().toString()
                .equals("Delete")) {

            subjects.remove(selectedPosition);

            // Refresh ListView after deleting
            adapter.notifyDataSetChanged();

            Toast.makeText(
                    this,
                    "Deleted " + selectedSubject,
                    Toast.LENGTH_SHORT
            ).show();

            selectedPosition = -1;

            return true;
        }

        // If Open Website was selected
        if (item.getTitle().toString()
                .equals("Open Website")) {

            // ----------------------------------------
            // IMPLICIT INTENT
            // ----------------------------------------

            Intent intent = new Intent(
                    Intent.ACTION_VIEW,
                    Uri.parse("https://www.google.com")
            );

            // Check whether an app can handle the Intent
            if (intent.resolveActivity(
                    getPackageManager()) != null) {

                startActivity(intent);
            }

            return true;
        }

        return super.onContextItemSelected(item);
    }
}
```

8. MANIFEST

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <application
        android:label="Simple MAD">

        <!-- Second Activity -->
        <activity
            android:name=".HomeActivity"
            android:exported="false" />

        <!-- First Activity / Launcher -->
        <activity
            android:name=".MainActivity"
            android:exported="true">

            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category
                    android:name="android.intent.category.LAUNCHER" />
            </intent-filter>

        </activity>

    </application>

</manifest>
```

9. THE SAME PROGRAM AS A MEMORY MAP

```
MAIN ACTIVITY
    |
    |-- findViewById()
    |
    |-- Read name/course
    |
    |-- Validate
    |      |
    |      +-- setError()
    |
    |-- RadioButton
    |
    |-- CheckBox
    |
    |-- SharedPreferences
    |
    |-- AlertDialog
    |
    |-- Bundle
    |
    +-- Explicit Intent
            |
            v
       HOME ACTIVITY
            |
            |-- getIntent().getExtras()
            |
            |-- ListView
            |      |
            |      +-- ArrayAdapter
            |
            |-- Normal click
            |      |
            |      +-- AlertDialog + Toast
            |
            |-- Long click
            |      |
            |      +-- Context Menu
            |              |
            |              +-- Delete
            |              |     |
            |              |     +-- notifyDataSetChanged()
            |              |
            |              +-- Open Website
            |                     |
            |                     +-- Implicit Intent
            |
            v
          END
```

10. HOW TO CHANGE THIS FOR ANY EXAM QUESTION

If question asks for...	Change this example to...
Patient appointment	name → patientName; course → appointmentType; add date/time.
Shopping cart	subjects → products; CheckBoxes → products; calculate total.
Contact list	subjects → contacts; Context Menu → Call/Edit/Delete.
Travel planner	subjects → destinations; Implicit Intent → Maps.
Email utility	Open Website → ACTION_SENDTO mailto.
Phone utility	Open Website → ACTION_DIAL tel:.
SMS utility	Open Website → ACTION_SENDTO sms:.
Calendar	Use ACTION_INSERT for calendar event.
Fragments	Add FrameLayout, create Fragment, replace(), addToBackStack(), setArguments()/getArguments().

11. MOST IMPORTANT LINES TO MEMORIZE

```
// 1. Read EditText
String name = nameEditText.getText().toString().trim();

// 2. Validate
if (name.isEmpty()) {
    nameEditText.setError("Enter name");
    return;
}

// 3. Toast
Toast.makeText(this, "Saved", Toast.LENGTH_SHORT).show();

// 4. Save preference
preferences.edit().putString("studentName", name).apply();

// 5. Read preference
String name = preferences.getString("studentName", "");

// 6. Explicit Intent
Intent intent = new Intent(MainActivity.this, HomeActivity.class);
startActivity(intent);

// 7. Send data
intent.putExtra("name", name);

// 8. Receive data
String name = getIntent().getStringExtra("name");

// 9. ListView adapter
ArrayAdapter<String> adapter =
        new ArrayAdapter<>(this,
                android.R.layout.simple_list_item_1,
                subjects);

subjectListView.setAdapter(adapter);

// 10. Refresh after delete
adapter.notifyDataSetChanged();

// 11. Implicit Intent
Intent intent = new Intent(
        Intent.ACTION_VIEW,
        Uri.parse("https://www.google.com")
);

if (intent.resolveActivity(getPackageManager()) != null) {
    startActivity(intent);
}
```

12. EXAM RULE TO REMEMBER

Do not try to memorize 50 completely different programs. Memorize the pattern. Then change the names, arrays, messages and destination according to the question.

Think: Input → Validate → Store → Confirm → Send → Display → Click → Long Click → Action.